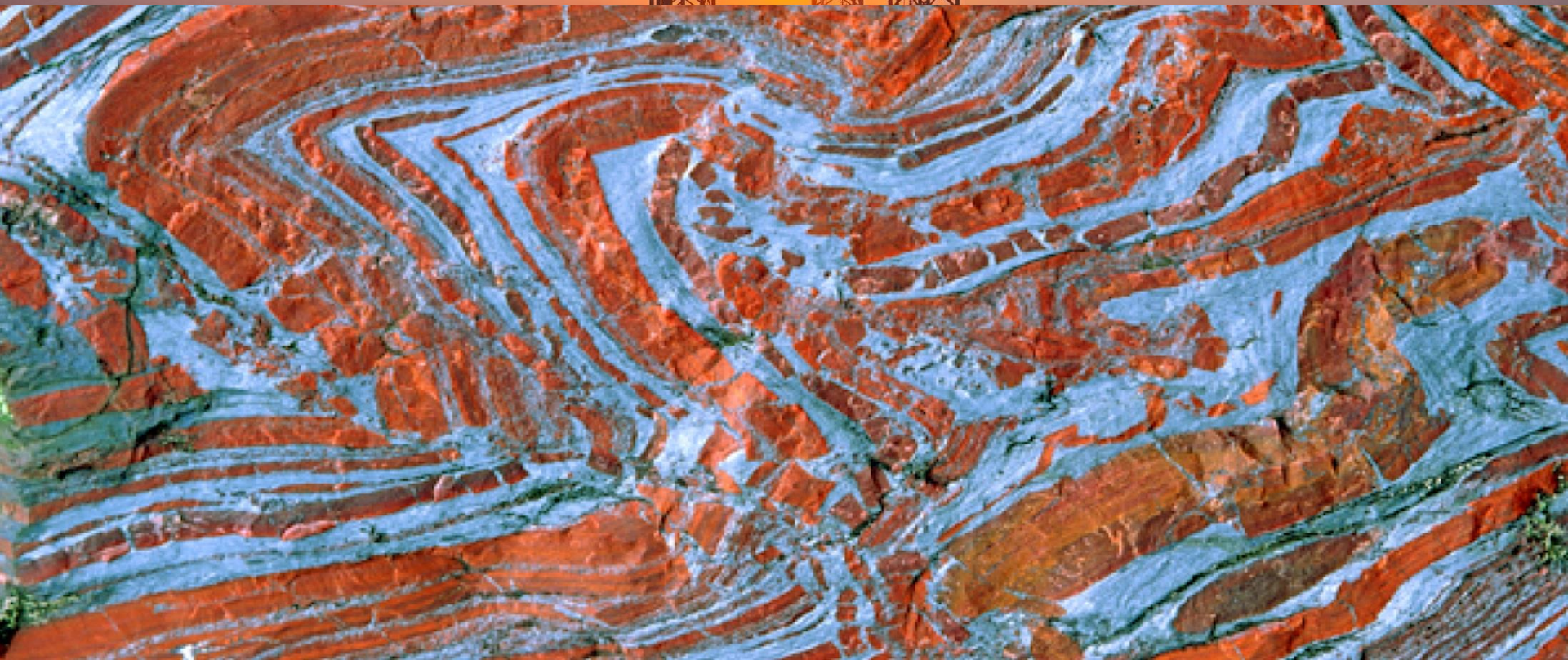
EART 26001
Energy and Resources

Dr Stefan Schroeder

Context for energy and resources



Energy and Resources
Learning goals week 1

This week you will learn to:

- **Describe what raw materials fall under energy and resources, and how they are important for current and future human development.**
- **Define the concept of critical raw materials (CRM).**
- **Define the concept of reserves and resources.**
- **Recognize the geological, economic, environmental, and societal constraints on resource identification and extraction.**
- **Describe some fundamental techniques and skills used in resource exploration and assessment.**

Energy and Resources in human society
Resource classification

Metals (incl precious *)	Non-metallic minerals (incl precious *)	Energy fuels	Renewable resources
Aluminium	Aggregate	Carbon (oil, gas, coal)	Hydrogen
Copper	Building & ornamental stones	Uranium	Solar
Gold *	Cement/lime		Biomass
Iron	Diamond *		Wind
Lithium	Graphite		Geothermal
Manganese	Gypsum, potash, salt		
Nickel	Phosphate		Water
PGE *	Silica		
REE	Sulfur		

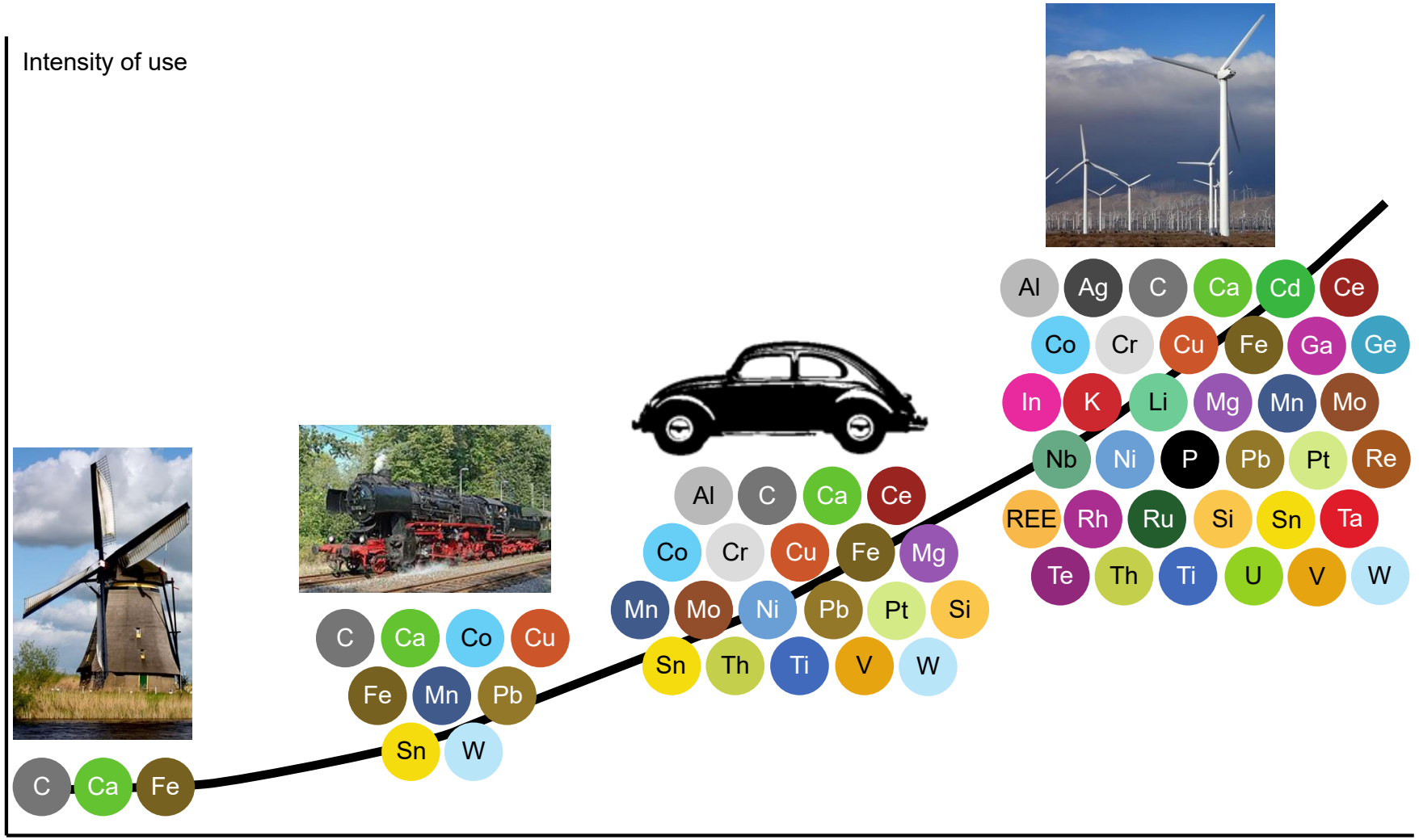
Key for energy transition

Energy and Resources in human society
Resources through time





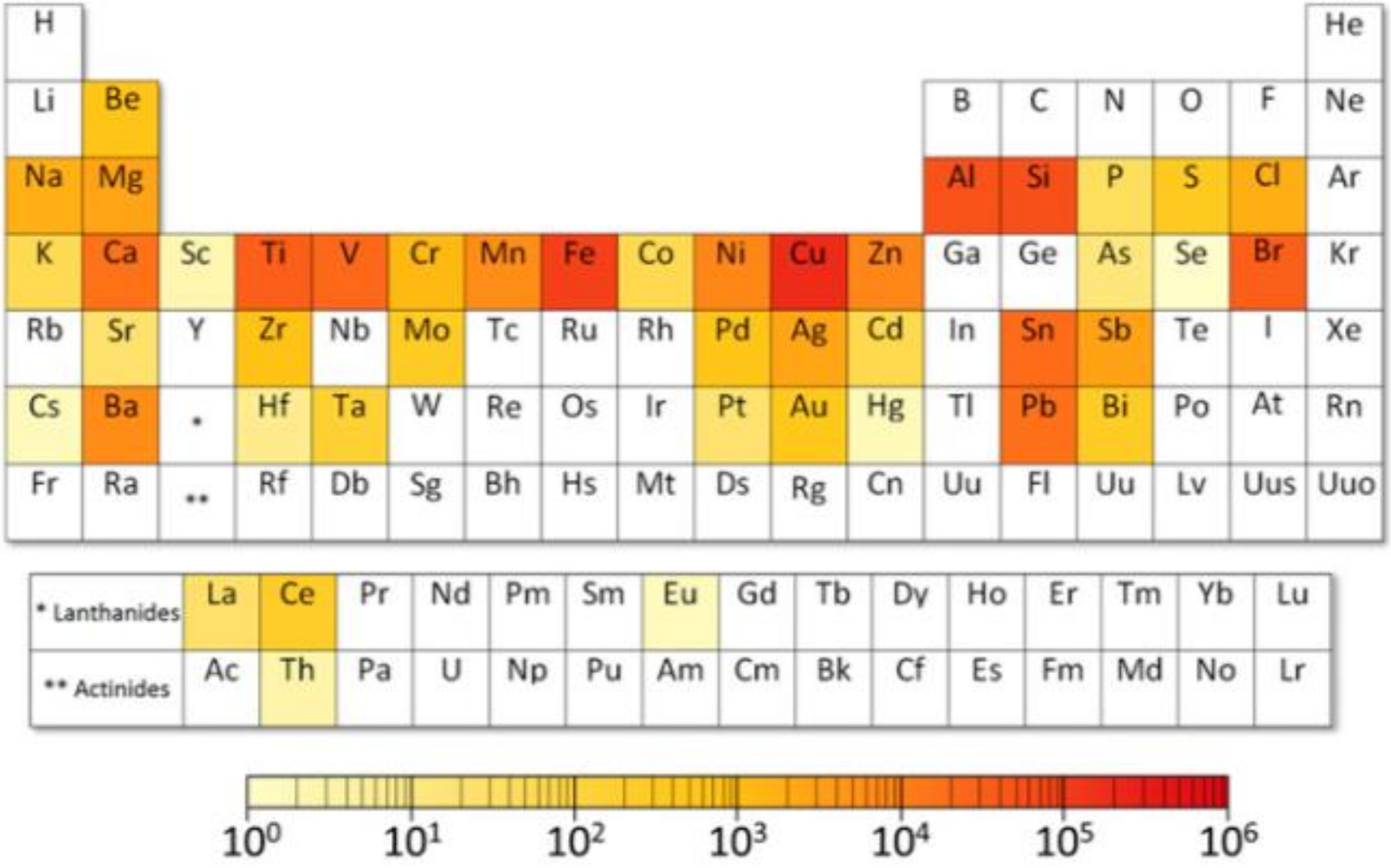
Energy and Resources in human society
Resources through time



After Achzet 2011

Energy and Resources in human society
Resources through time

Elements used in a computer circuit board



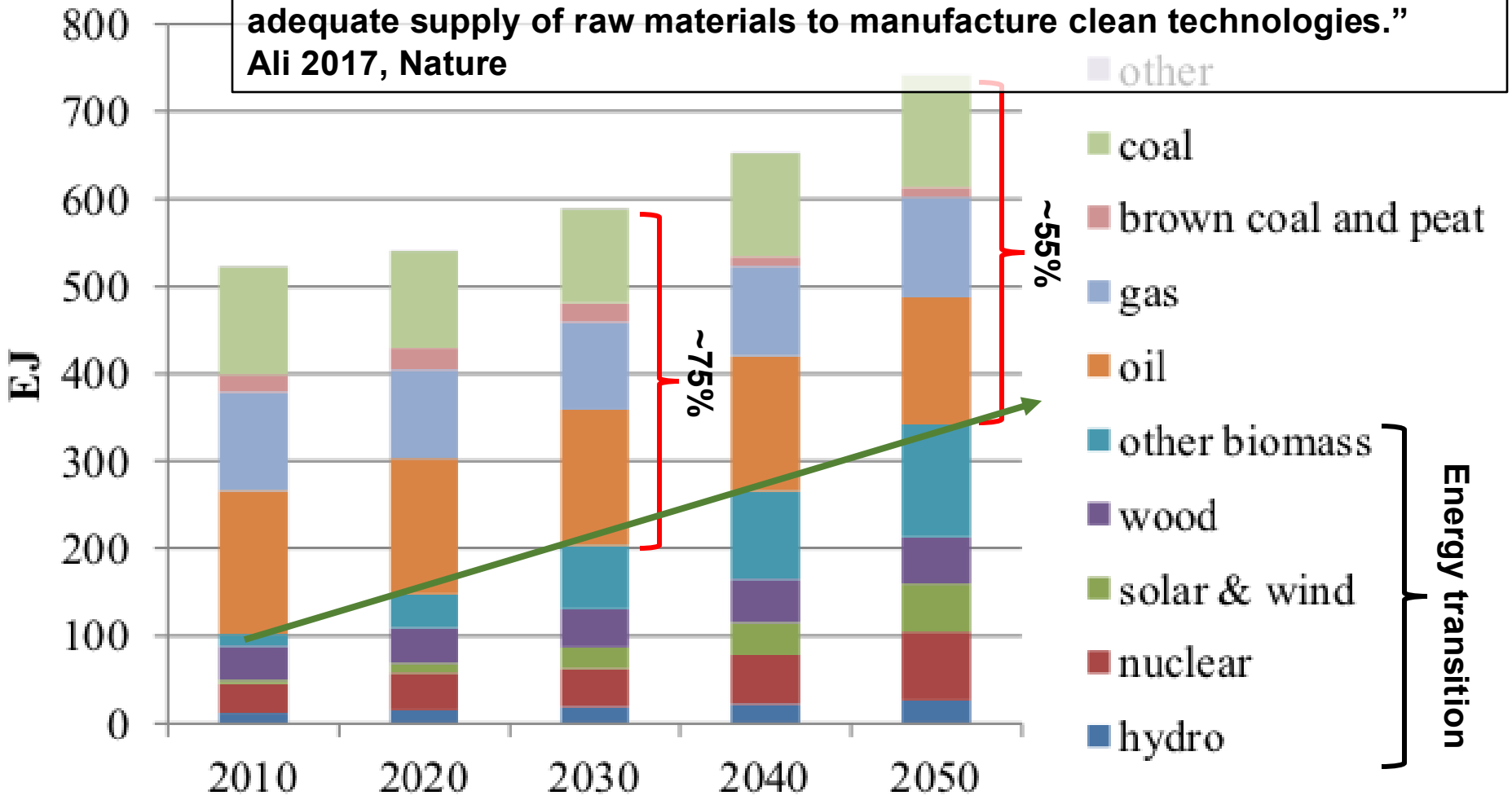
Concentration in ppm

Graedel 2015

Energy and Resources in human society

Future demand for resources - energy

"A transition to a low-carbon society….will require vast amounts of metals and minerals….because the world cannot tackle climate change without an adequate supply of raw materials to manufacture clean technologies."
Ali 2017, Nature



Grandell 2016

Energy and Resources in human society

Future demand for resources - minerals





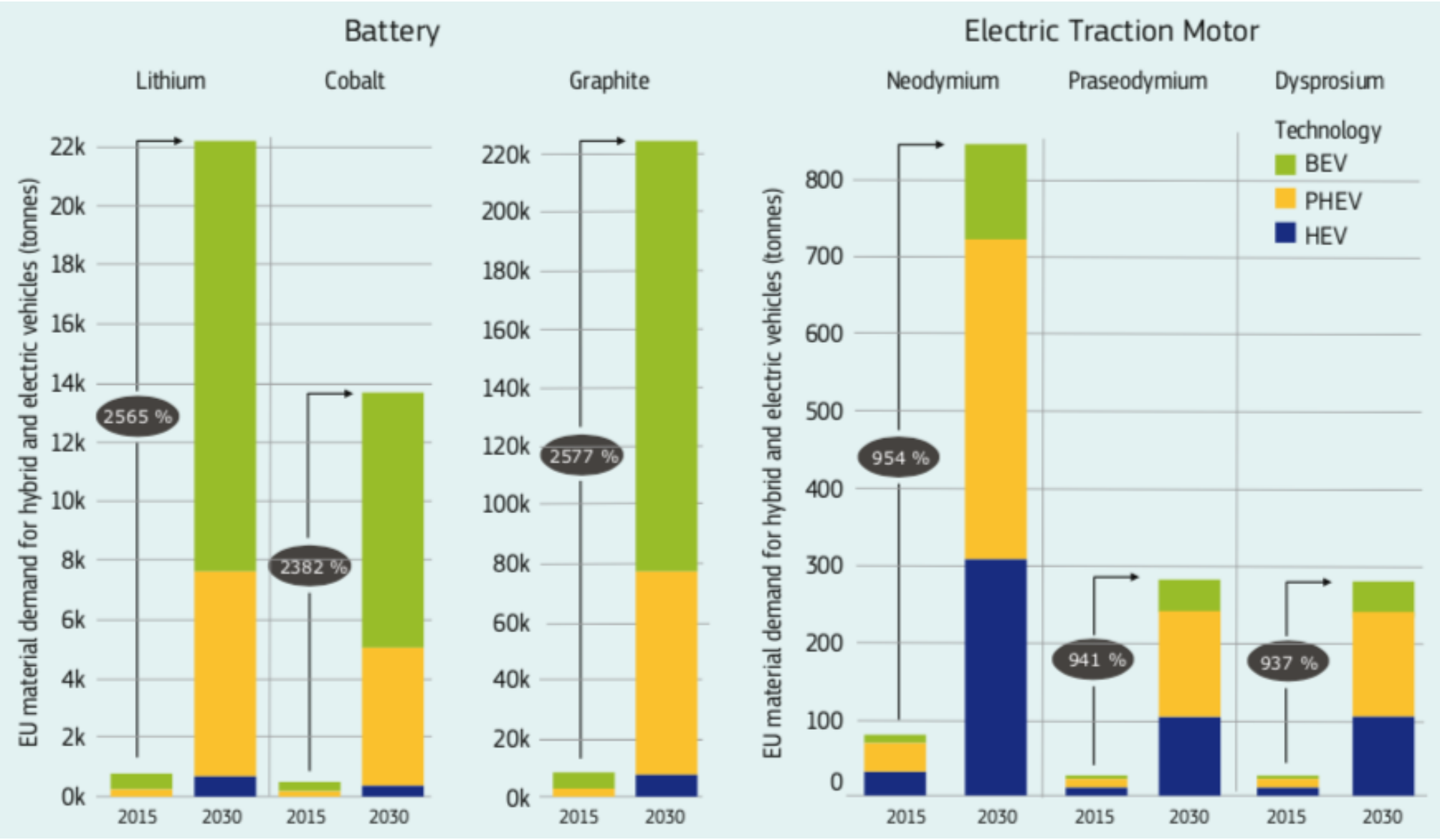


Critical mineral needs for clean energy technologies

Source: IEA

	Copper	Cobalt	Nickel	Lithium	REEs	Chromium	Zinc	PGMs	Aluminium*
Solar PV	●	○	○	○	○	○	○	○	●
Wind	●	○	◍	○	●	◍	●	○	◍
Hydro	◍	○	○	○	○	◍	◍	○	◍
CSP	◍	○	◍	○	○	●	◍	○	●
Bioenergy	●	○	○	○	○	○	◍	○	◍
Geothermal	○	○	●	○	○	●	○	○	○
Nuclear	◍	○	◍	○	○	◍	○	○	○
Electricity networks	●	○	○	○	○	○	○	○	●
EVs and battery storage	●	●	●	●	●	○	○	○	●
Hydrogen	○	○	●	○	◍	○	○	●	◍

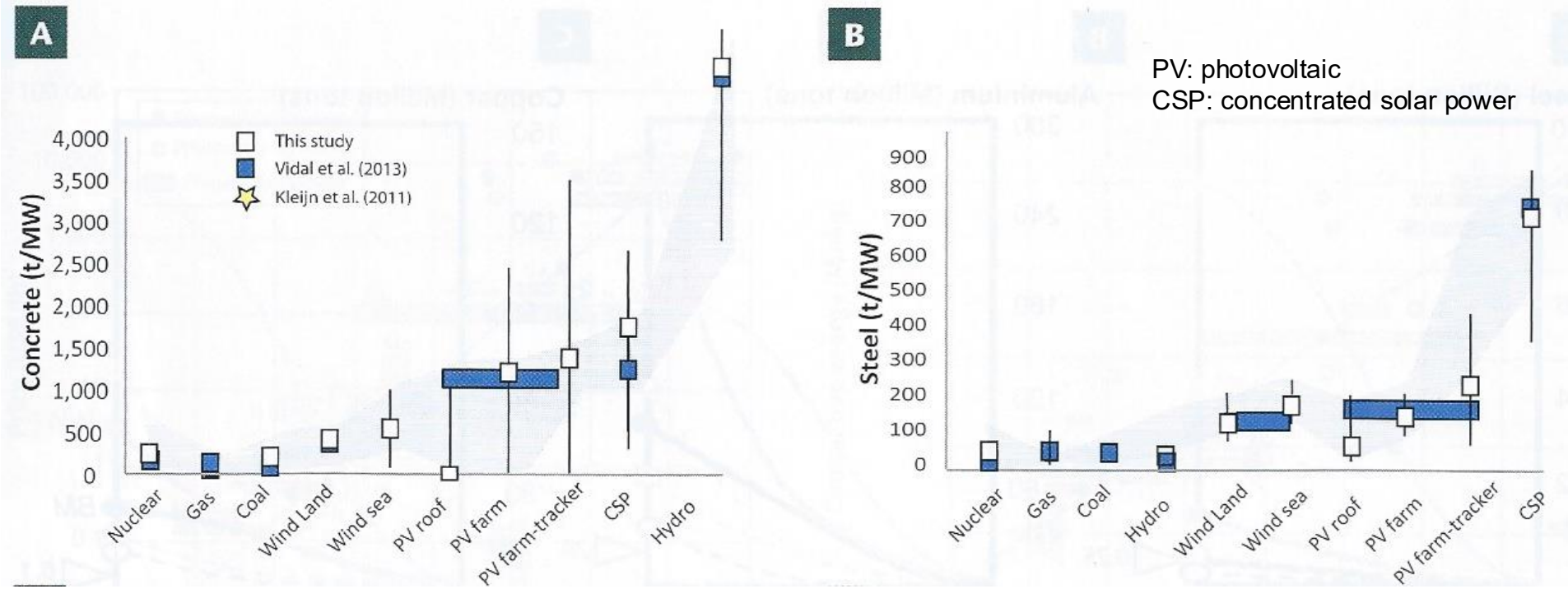
Energy and Resources in human society
Future demand for resources - minerals



European Commission Raw Materials Scoreboard 2018

Energy and Resources in human society
Future demand for resources - minerals

Material needs per MW generated, for different energy sources



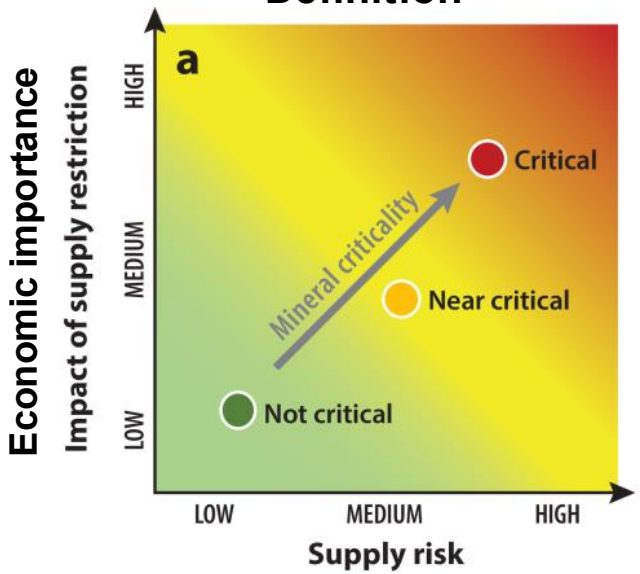
PV: photovoltaic
CSP: concentrated solar power

Vidal 2017

- **Increasing demand in metals and non-minerals for high-tech materials and for infrastructure**

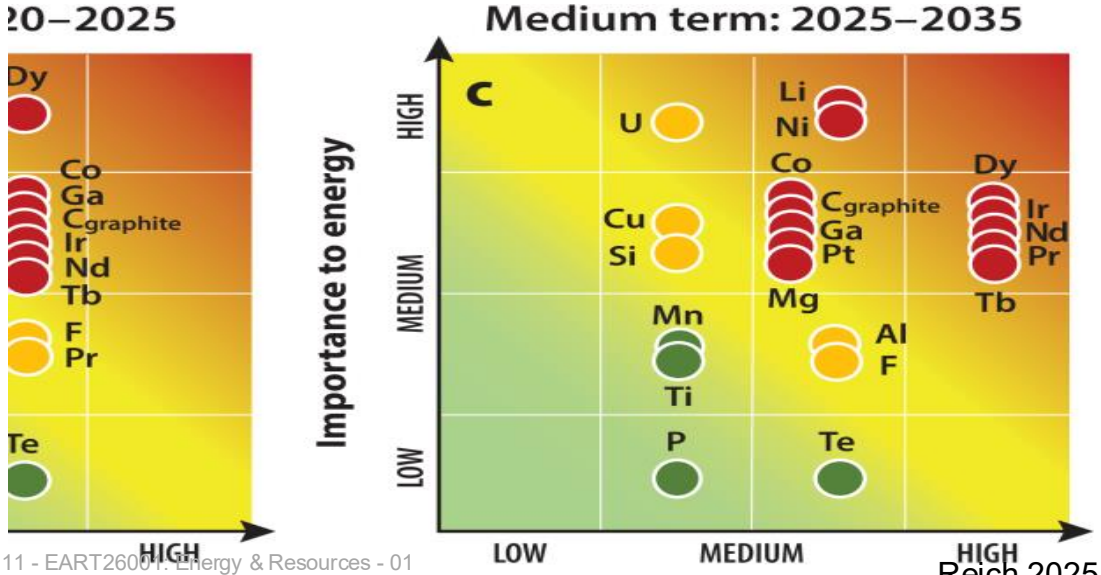
Energy and Resources in human society
Future demand for resources – critical resources

Definition



- Elements/minerals/materials – overarching: critical raw materials (CRM)
- Assessed on basis of company, industry sector, country, economic bloc
- Assessment specific to stakeholders, fluctuates over time
- Take into account environmental and/or ethical issues

US Dept of Energy 2023 assessment



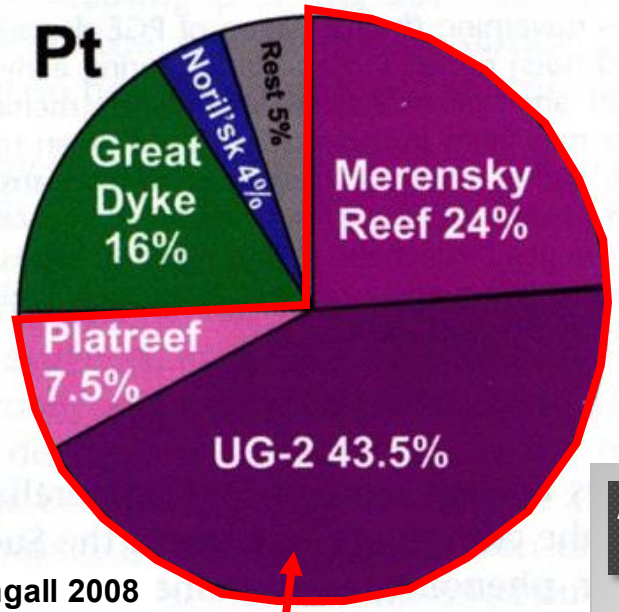
Reasons for criticality:

- Availability
- Production as a by-product of other processes
- Demand for high-tech consumer products
- Very low substitutability
- Virtually no recycling

Reich 2025

Energy and Resources in human society
Future demand for resources – critical resources (supply)

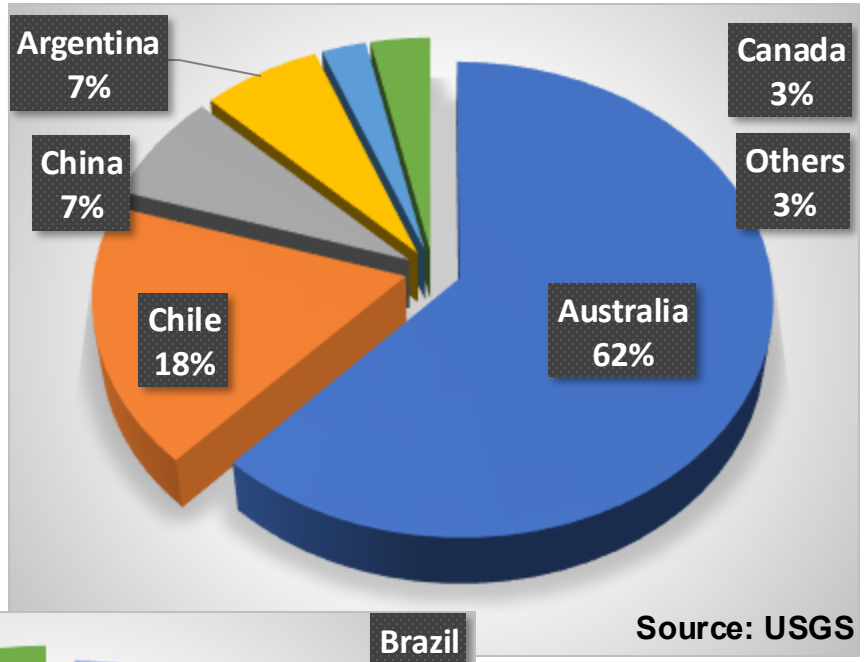
Share of deposits/countries for key metals



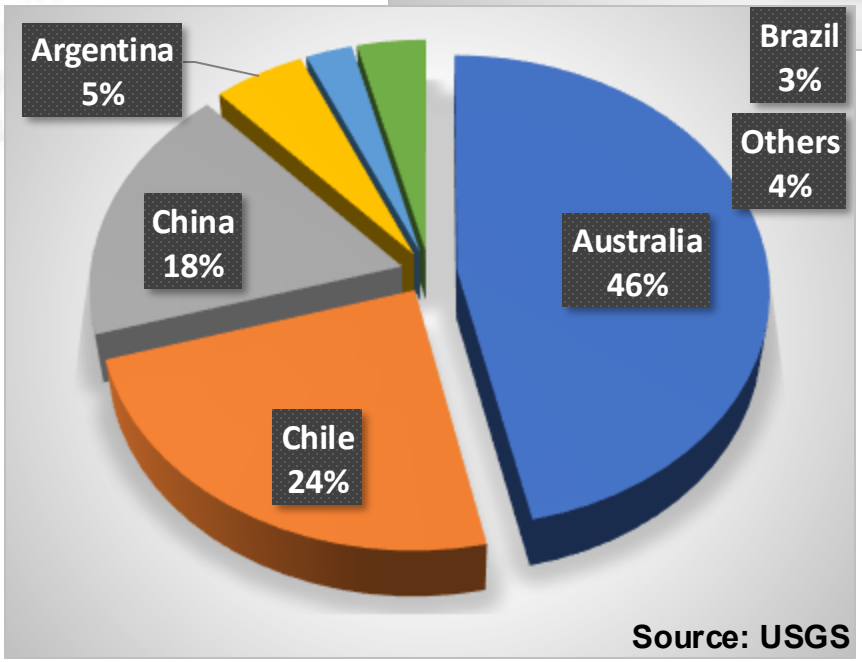
Mungall 2008

75% from single deposit in South Africa

Lithium mine production 2018....



Source: USGS



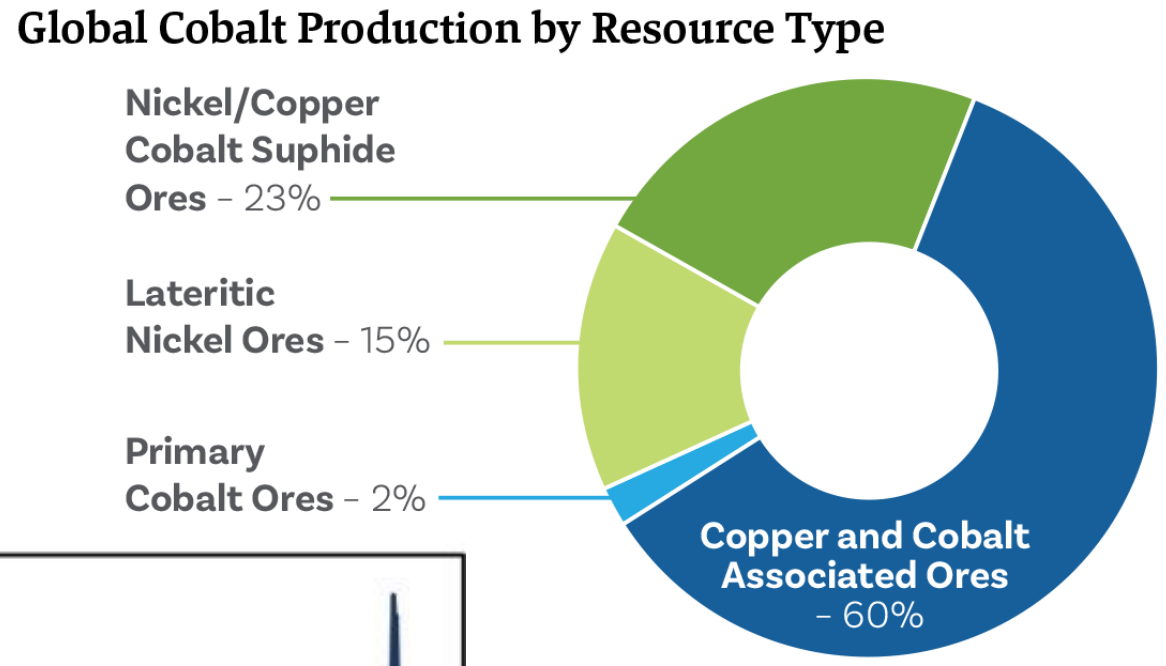
Source: USGS

....and 2023

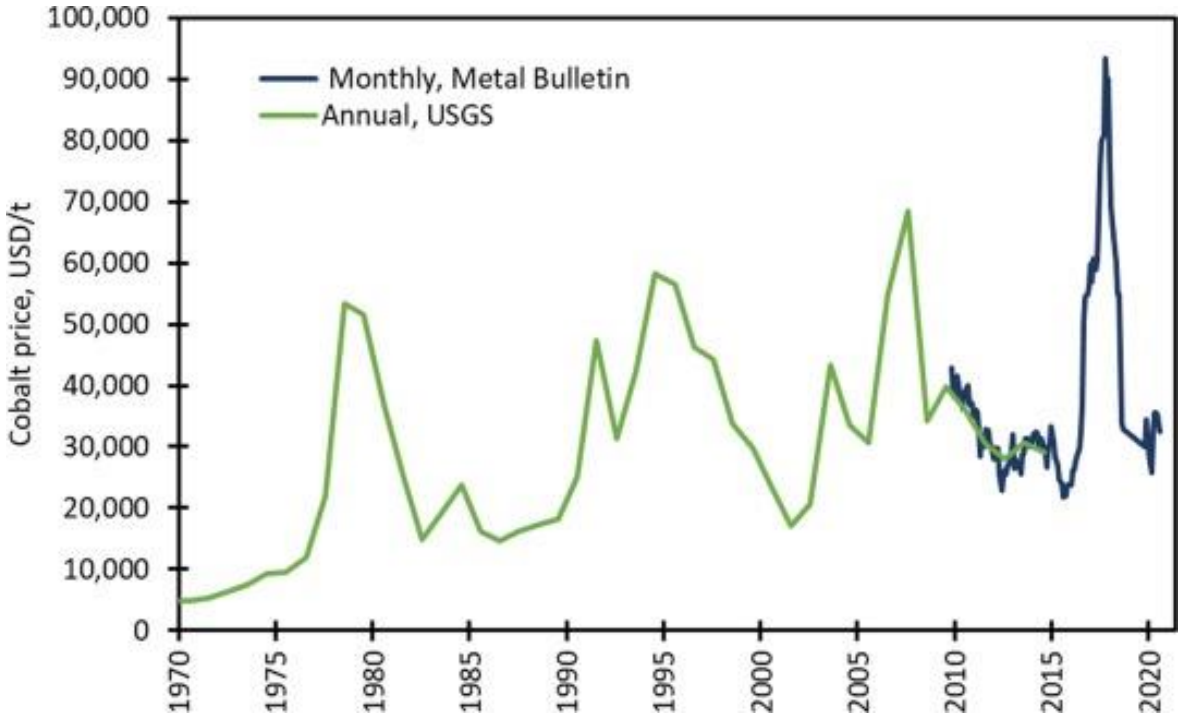
Energy and Resources in human society

Future demand for resources – critical resources (by-product)

68% mined in DRC (2020 production)

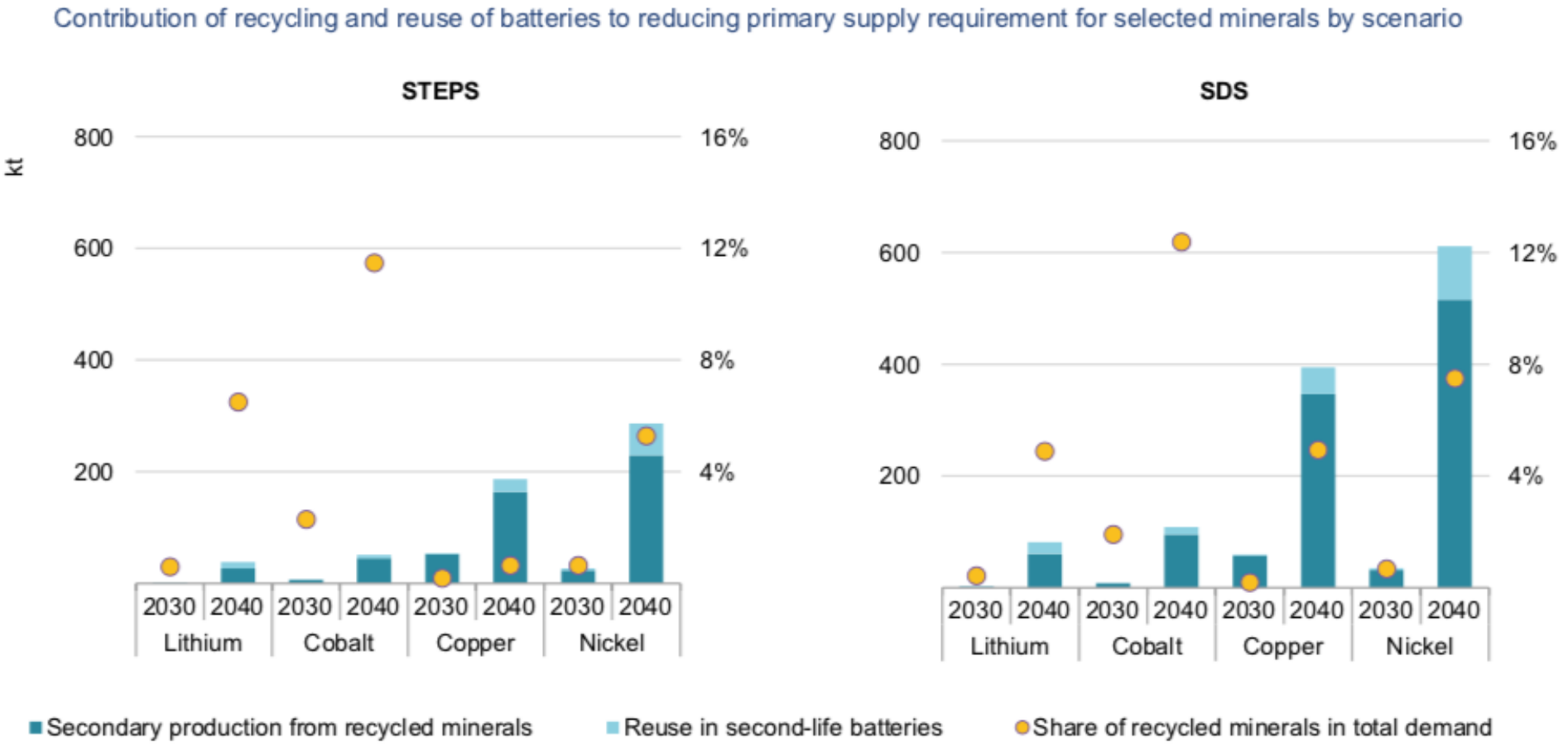


Source: USGS, Crux Investor



Crundwell 2020

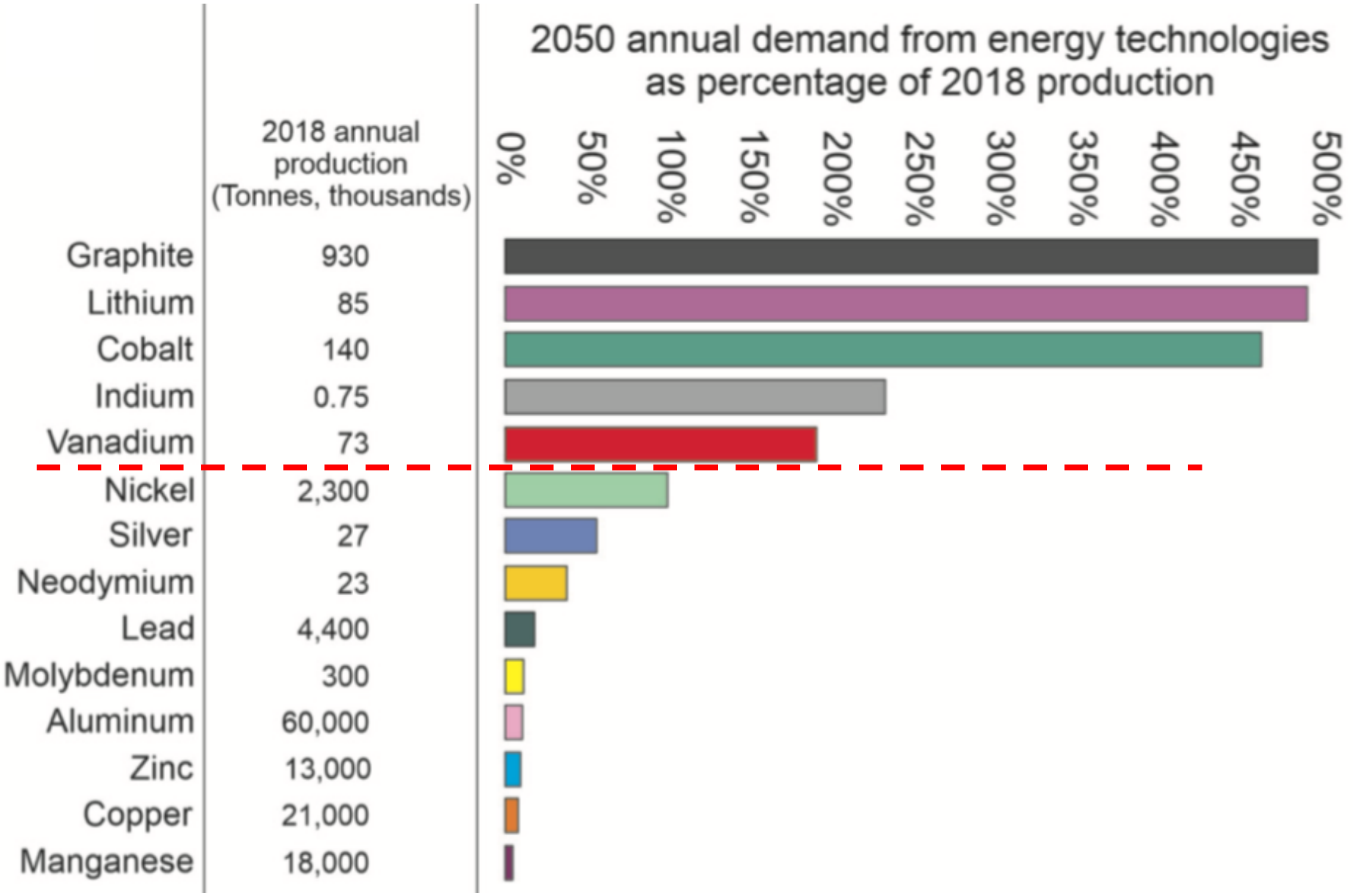
Energy and Resources in human society
Future demand for resources - critical resources (recycling)



Source: IEA 2022: The Role of Critical Minerals in Clean Energy Transitions

=> Continued demand for primary mining (and geoscientists…)

Energy and Resources in human society
Future demand for resources – do we have enough?

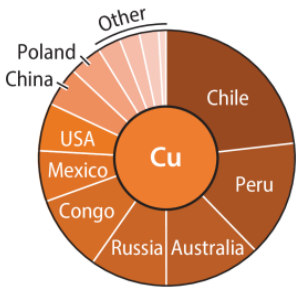


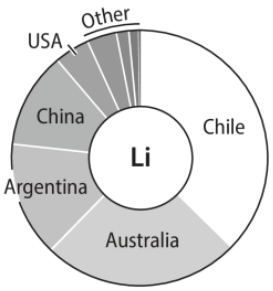
Simandl 2021

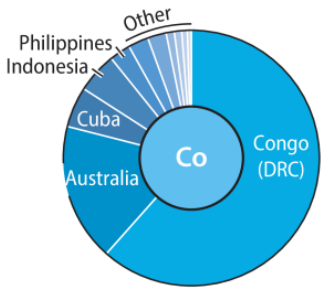
Energy and Resources in human society
Future demand for resources – do we have enough?

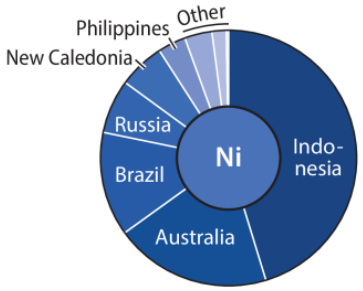
Global critical mineral deposits

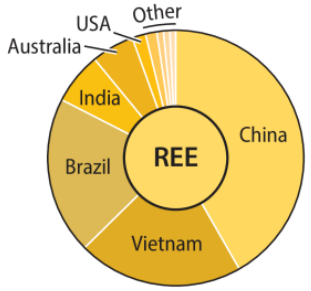










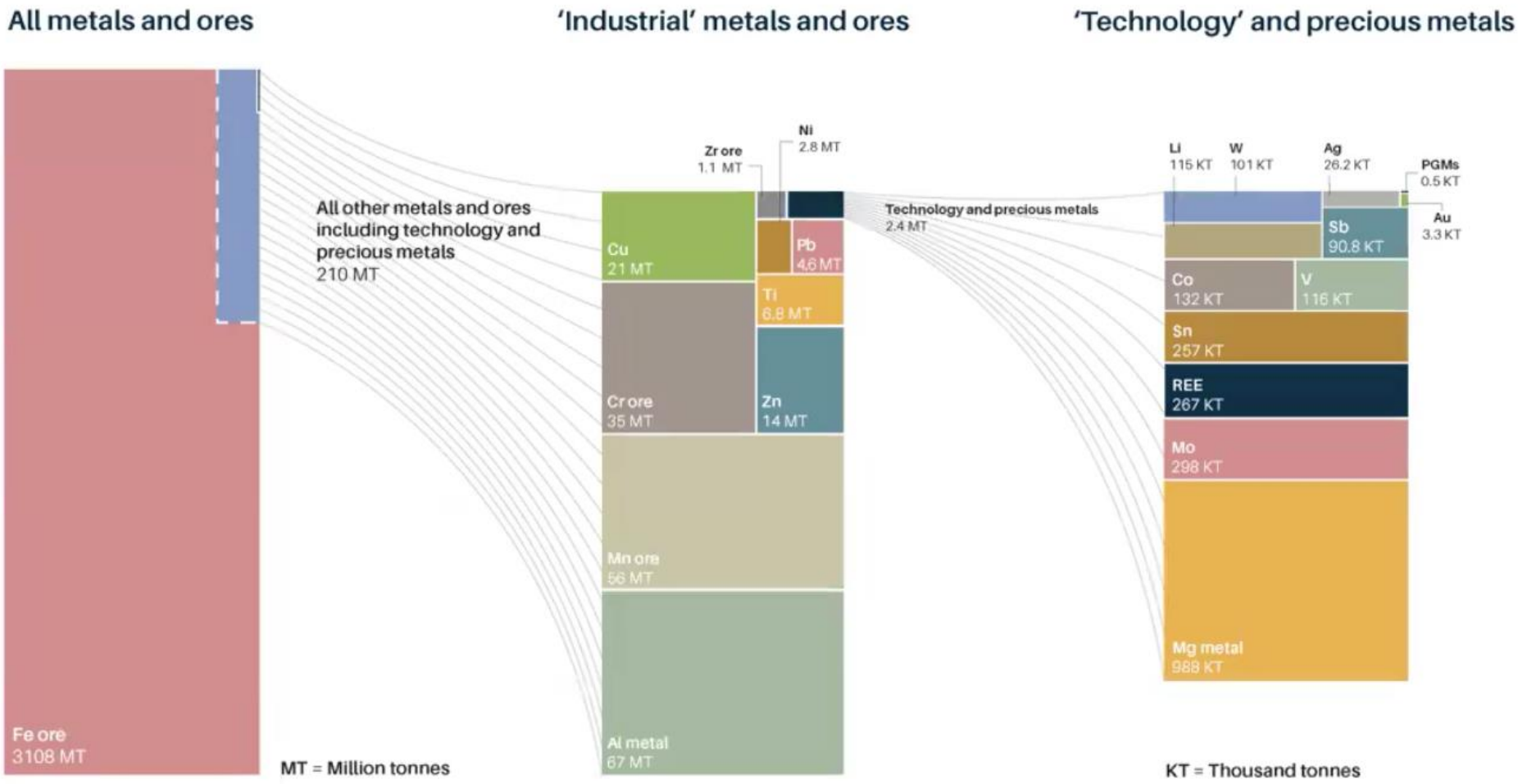


Reserves

Reich 2025, source: USGS 2024

Energy and Resources in human society
Future demand for resources – do we have enough?

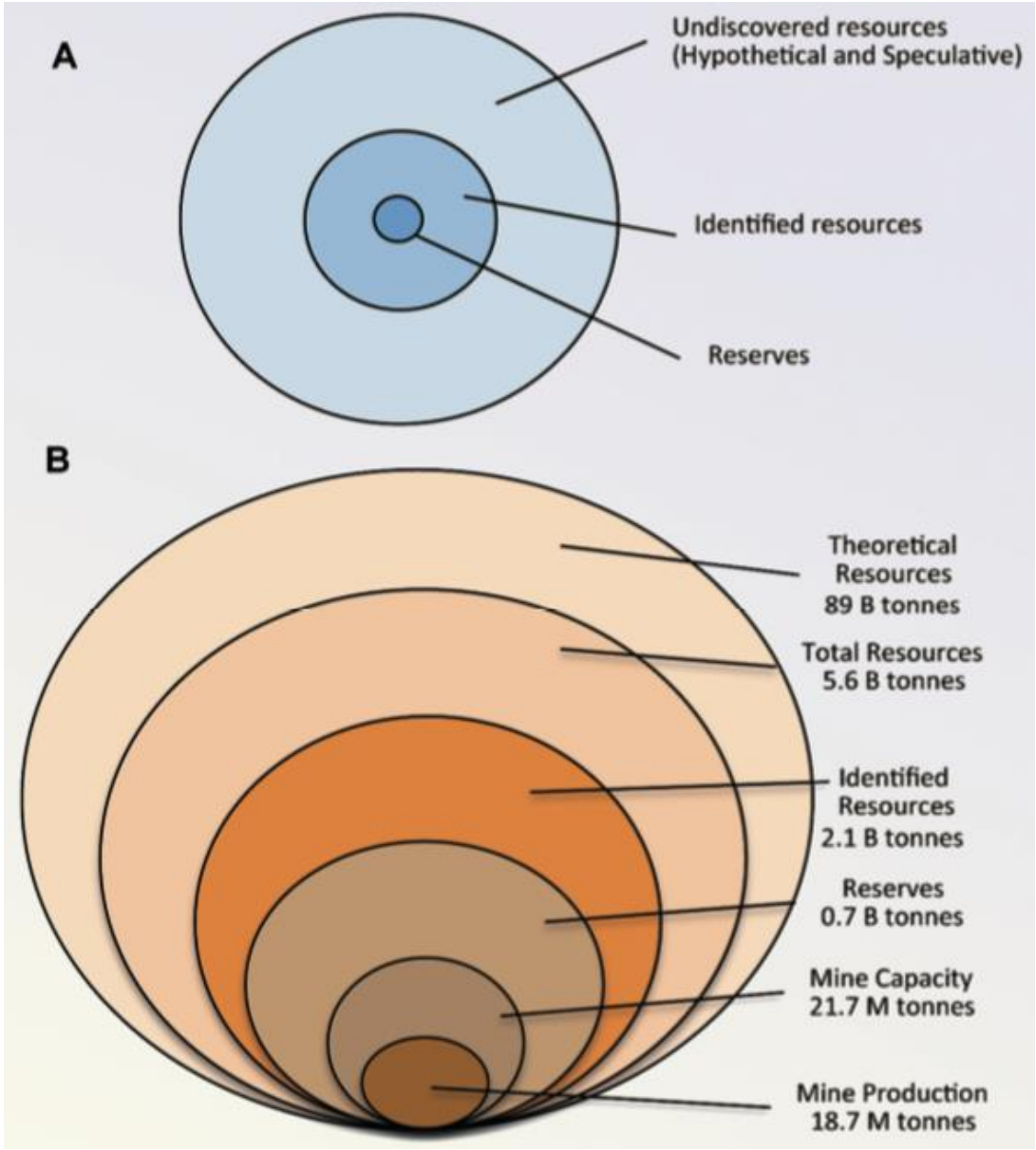
What do we mine?



Source: BGS

Energy and Resources – economic context
Resources and Reserves

- There is some confusion in reserve and resource estimates, and how they are interpreted
- Much of it hinges on definition of terms
- Reporting of reserves and reserves is legal requirement by government authorities and shareholders

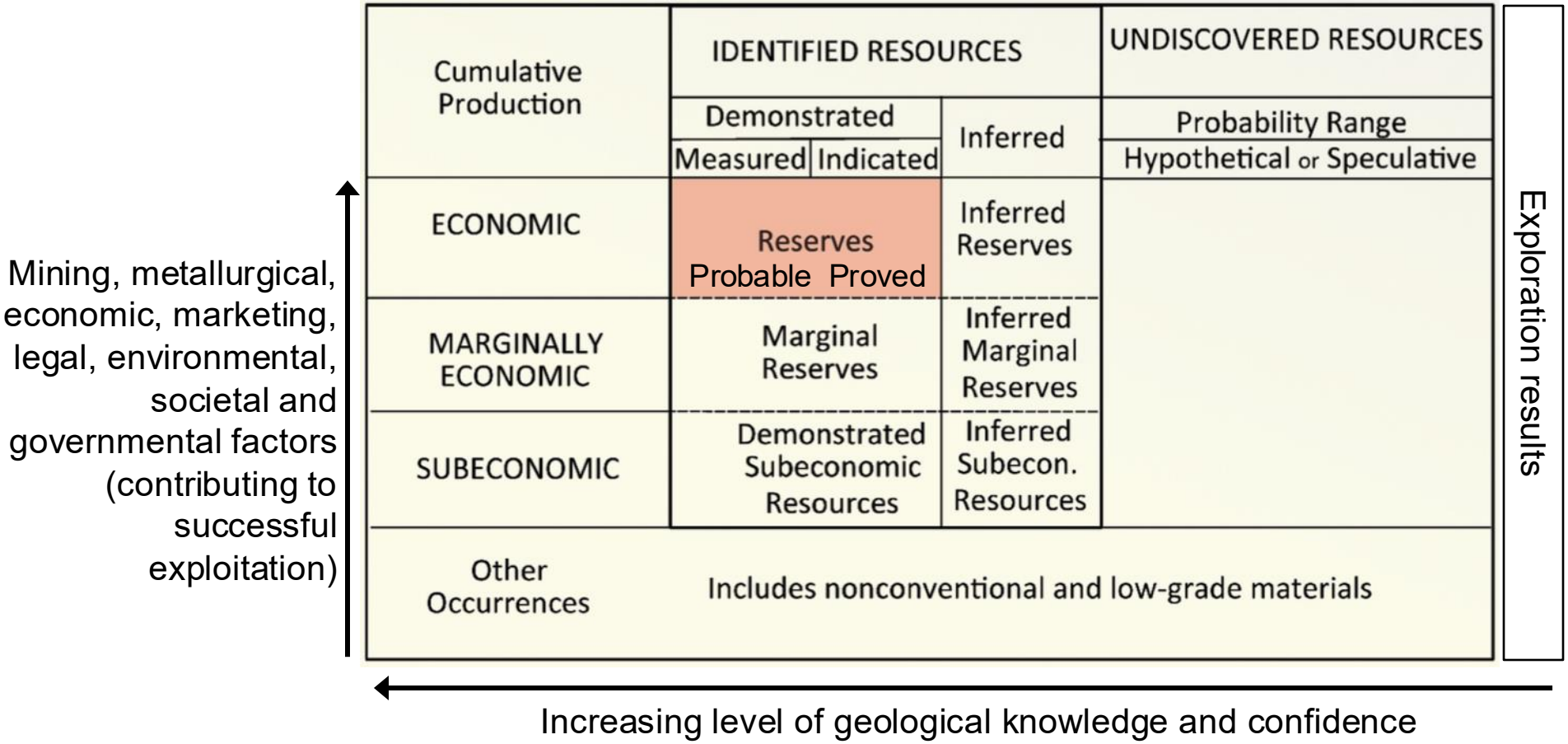


Arndt 2017

Energy and Resources – economic context

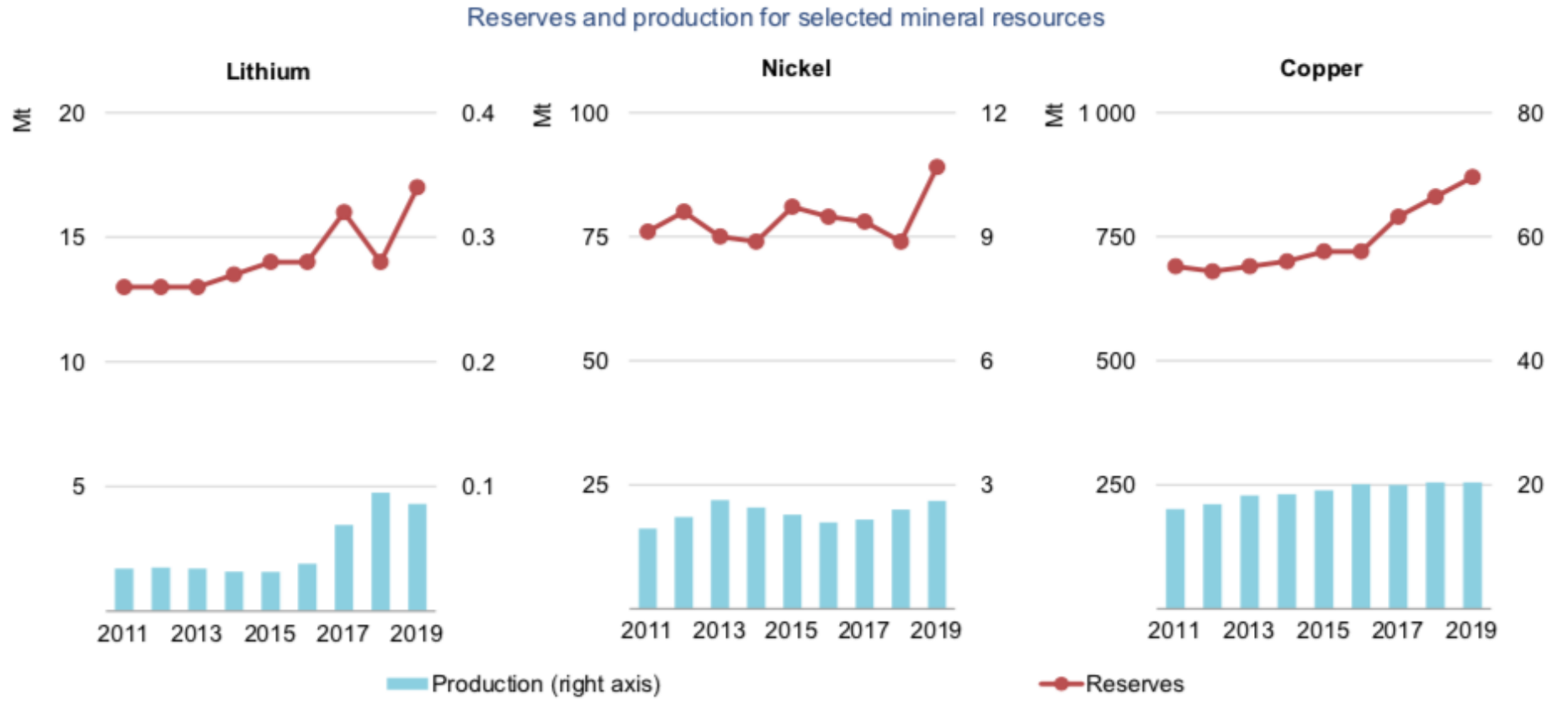
Resources and Reserves

Arndt 2017



- **Resources and reserves are dynamic entities that change over time depending on external factors**
- **Resource scarcity is not an issue**
- **Major issues are economic, environmental and social**

Energy and Resources – economic context
Resources and Reserves



IEA. All rights reserved.

Sources: USGS (2021); USGS (2020).

Source: IEA 2022: The Role of Critical Minerals in Clean Energy Transitions

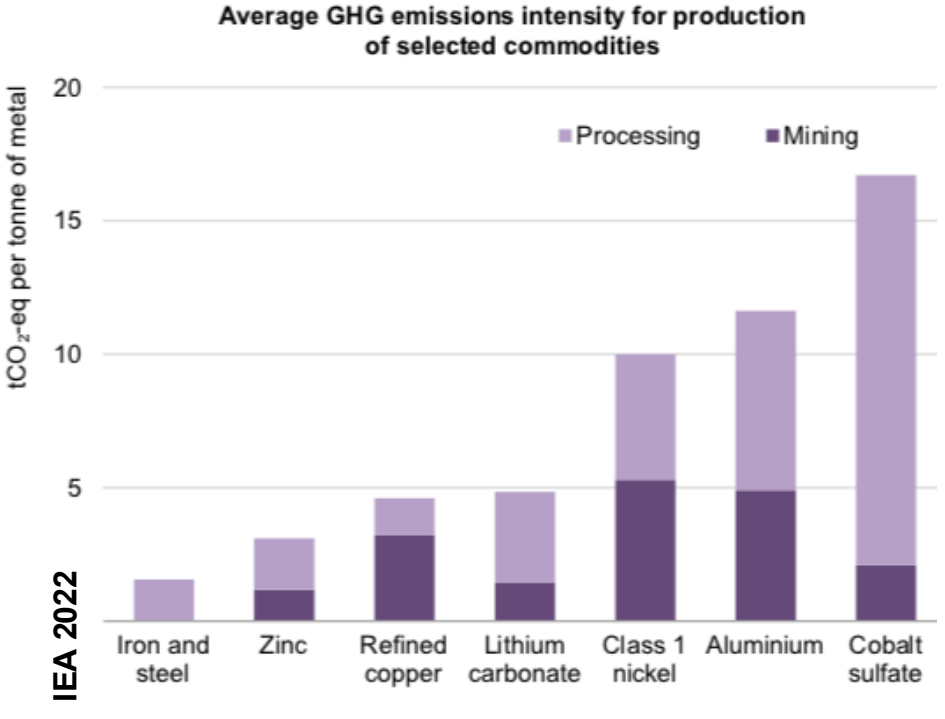
Energy and Resources in human society

Resource impact









Energy and Resources in human society
Future demand for resources – what to do, where to get resources?

- More sustainable consumption, more equality
- Mineral recycling ('urban mining'), circular economy -> less dependency on primary extraction
- Innovations in production and processing
  - Biomining and other processing techniques -> increased extraction efficiency
  - Re-processing of waste tailings -> increased extraction efficiency
  - Drilling vs strip mining, horizontal drilling -> lower footprint, increased extraction efficiency
  - Use tailings to offset CO_2 emissions
- Increased exploration and delivery of new resource projects (NB: geologists needed!)
- Use of big data, geochemistry and geophysics in exploration -> extending resource base through research (NB: geologists needed!)
- Understand resources – from geology to supply chains

Energy and Resources – Exploration & Production

"Resource deposits form when a useful commodity is sufficiently concentrated in an accessible part of the crust so it can be profitably extracted." (Robb 2008)

Cumulative Production	IDENTIFIED RESOURCES		UNDISCOVERED RESOURCES
	Demonstrated	Inferred	Probability Range
	Measured / Indicated		Hypothetical or Speculative
ECONOMIC	Reserves (Probable, Proved)	Inferred Reserves	
MARGINALLY ECONOMIC	Marginal Reserves	Inferred Marginal Reserves	
SUBECONOMIC	Demonstrated Subeconomic Resources	Inferred Subecon. Resources	
Other Occurrences	Includes nonconventional and low-grade materials		

Arndt 2017

Energy and Resources – Exploration & Production
Mineral grade

"Resource deposits form when a useful commodity is sufficiently concentrated in an accessible part of the crust so it can be profitably extracted." (Robb 2008)

Concentration of select metals in continental crust and ores

Clarke factor

Metal	Concentration in the crust[1]	Grade in ores[2]	Enrichment factor	Price (US$/kg)[3]
Al – aluminium	8 %	30 %	3-4	1.5
Fe – iron	4 %	50 %	10	0.1
Cu – copper	28 ppm	0.6 %	200	6
Nd – neodymium	27 ppm	1 %	300	100
U – uranium	2.7 ppm	0.2 – 2 %	100 – 1,000	80
Au – gold	1.5 ppb	6 ppm	4,000	50,000
Pt – platinum	0.5 ppb	5 ppm	10,000	50,000

[1] Upper continental crust from Rudnick and Gao (2004); Al and Fe recalculated from oxide values.
[2] From InfoMine.com.
[3] Typical price 2012-2016, from InfoMine.com.

Arndt 2017

Energy and Resources – Exploration & Production
Accessibility

"Resource deposits form when a useful commodity is sufficiently concentrated in an accessible part of the crust so it can be profitably extracted." (Robb 2008)





Source: Wikipedia

Deepest oil exploration well: 10.5 km (Tiber well, Gulf of Mexico) – but 2-5 km is standard
Deepest mine: 4 km (Mponeng gold mine, South Africa)

Energy and Resources – Exploration & Production
Uncertainty

Can you predict this…..



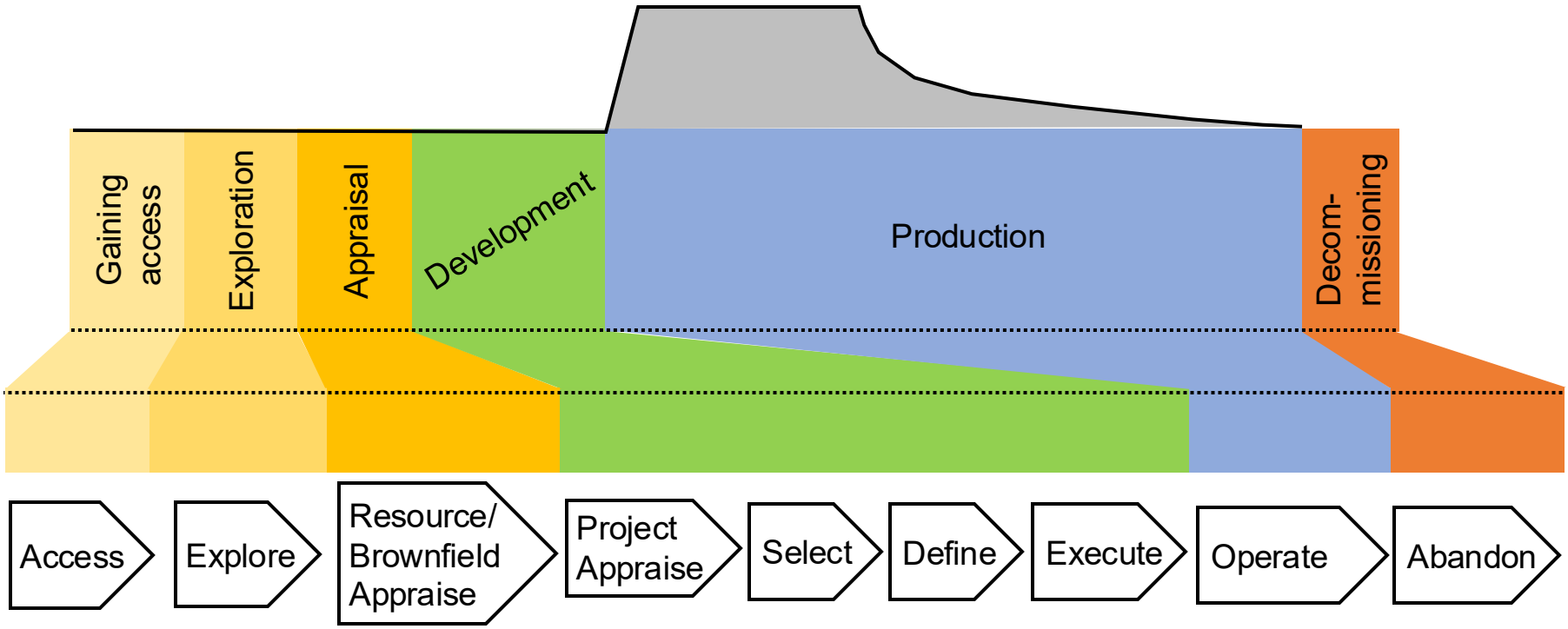
Energy and Resources – Exploration & Production
Uncertainty

…..from this…..?

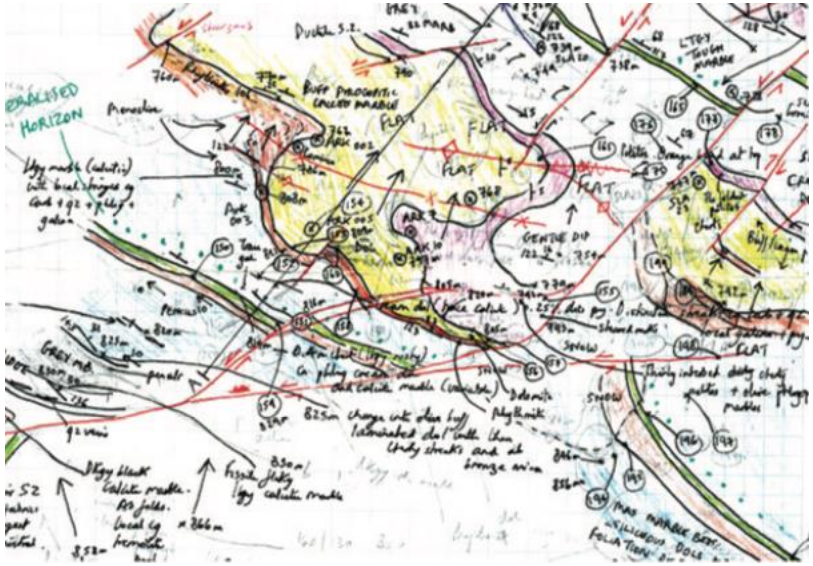


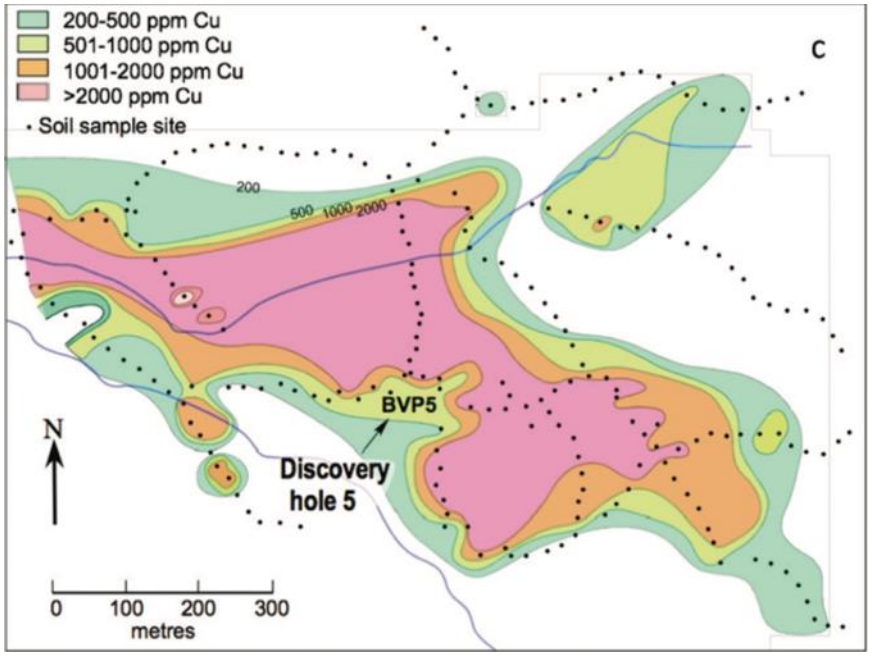
Energy and Resources – Exploration and Production

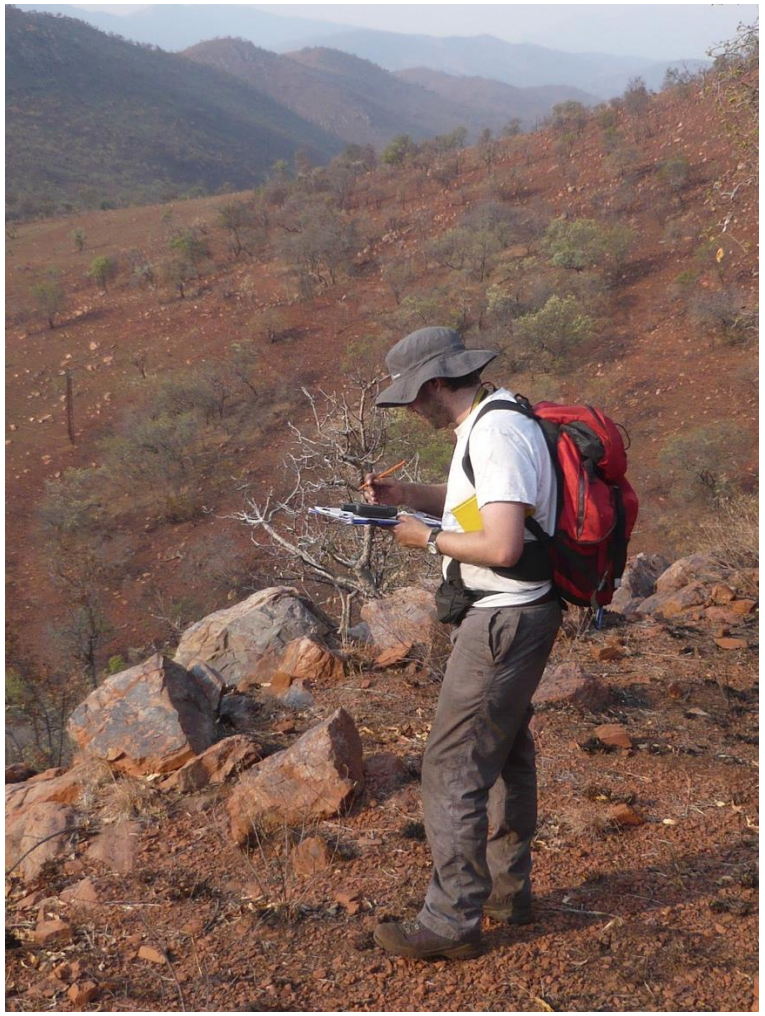
Resource Project Lifecycle



Energy and Resources – Exploration and Production
Exploration mapping



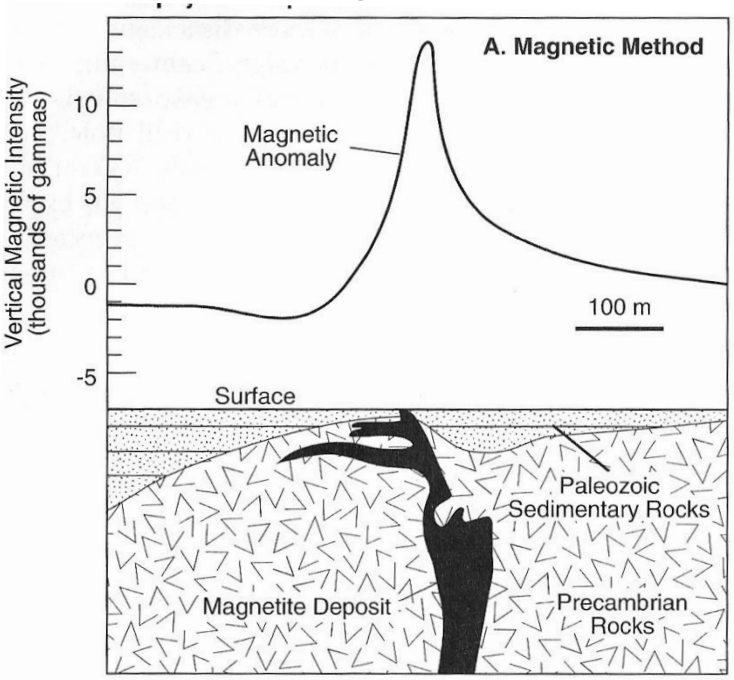




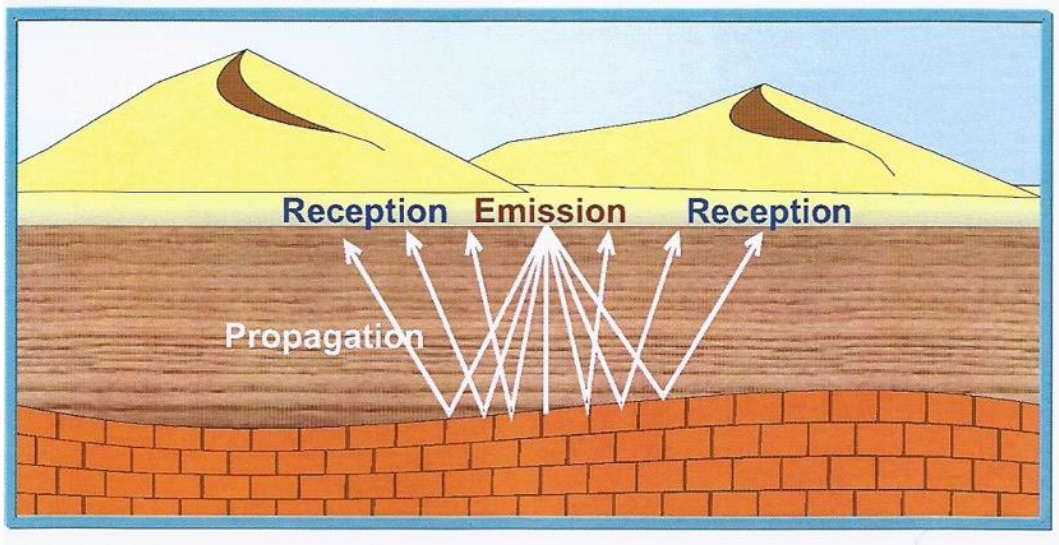
Arndt 2017

Energy and Resources – Exploration and Production
Exploration geophysics

Magnetics

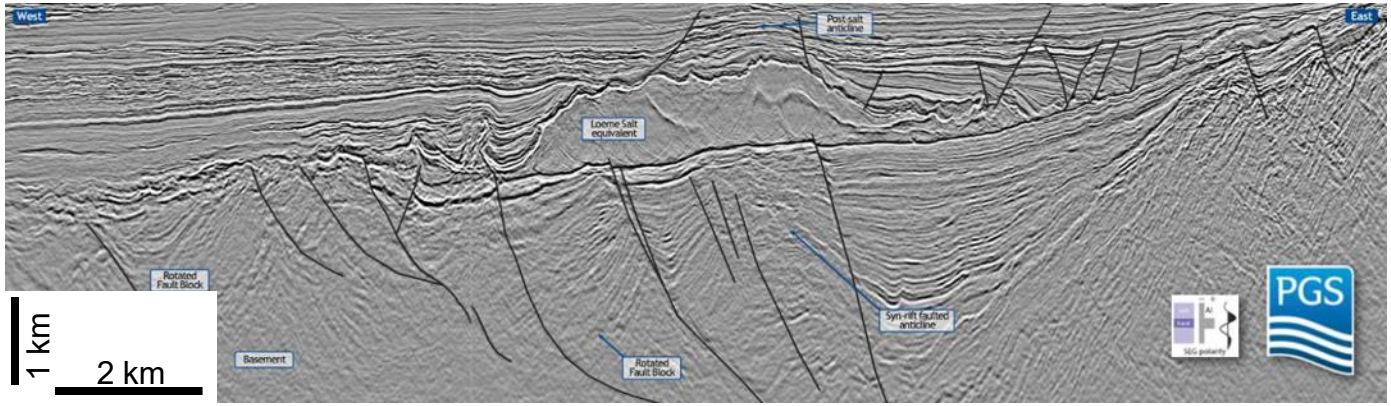


Seismic



Source: Total

Example seismic section of Angola



Reid 2016

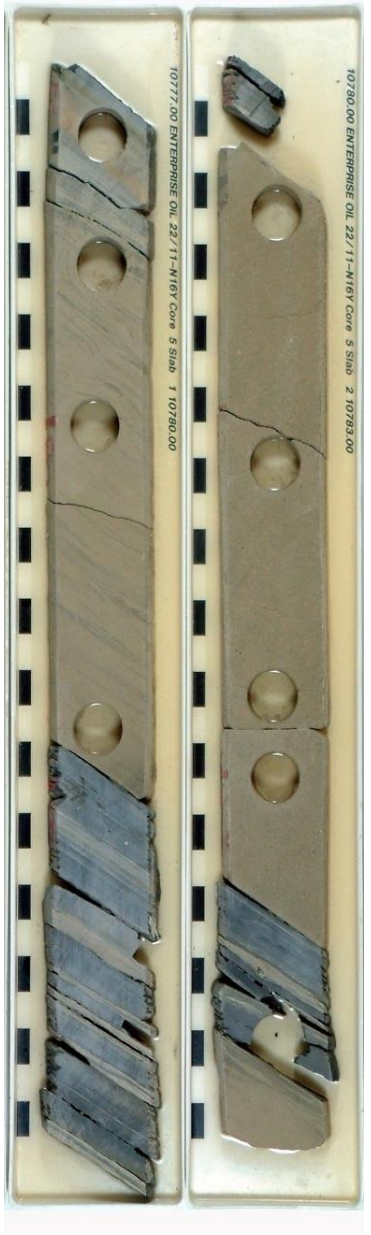
Energy and Resources – Exploration and Production
Drilling

Drilling and core logging





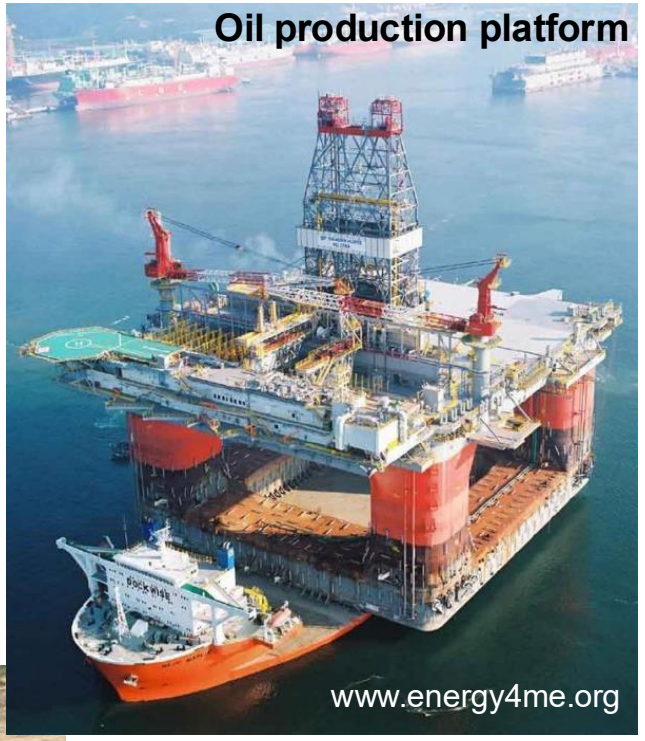




Energy and Resources – Exploration and Production
Production and processing



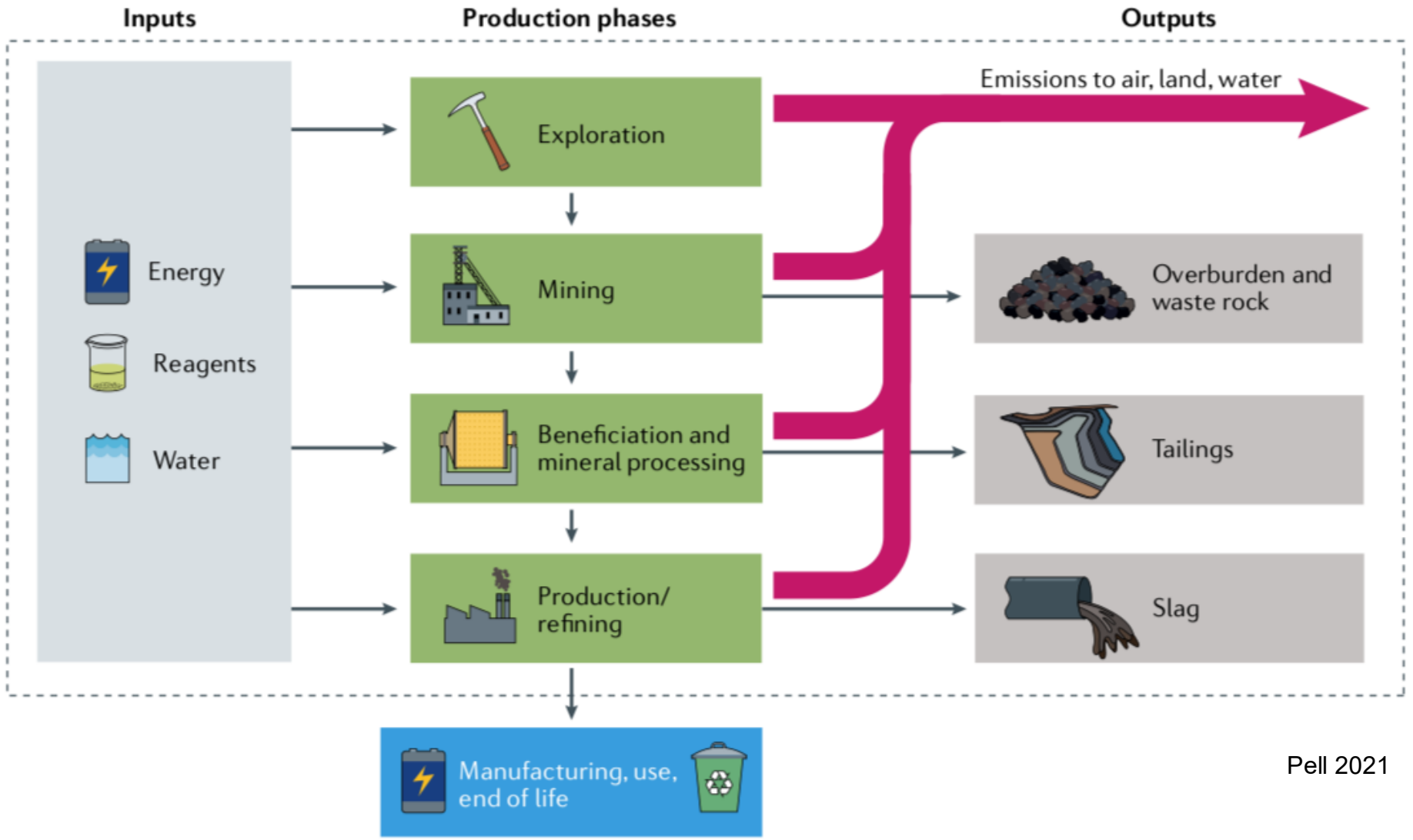
© Guardian





www.ftmmachinery.com

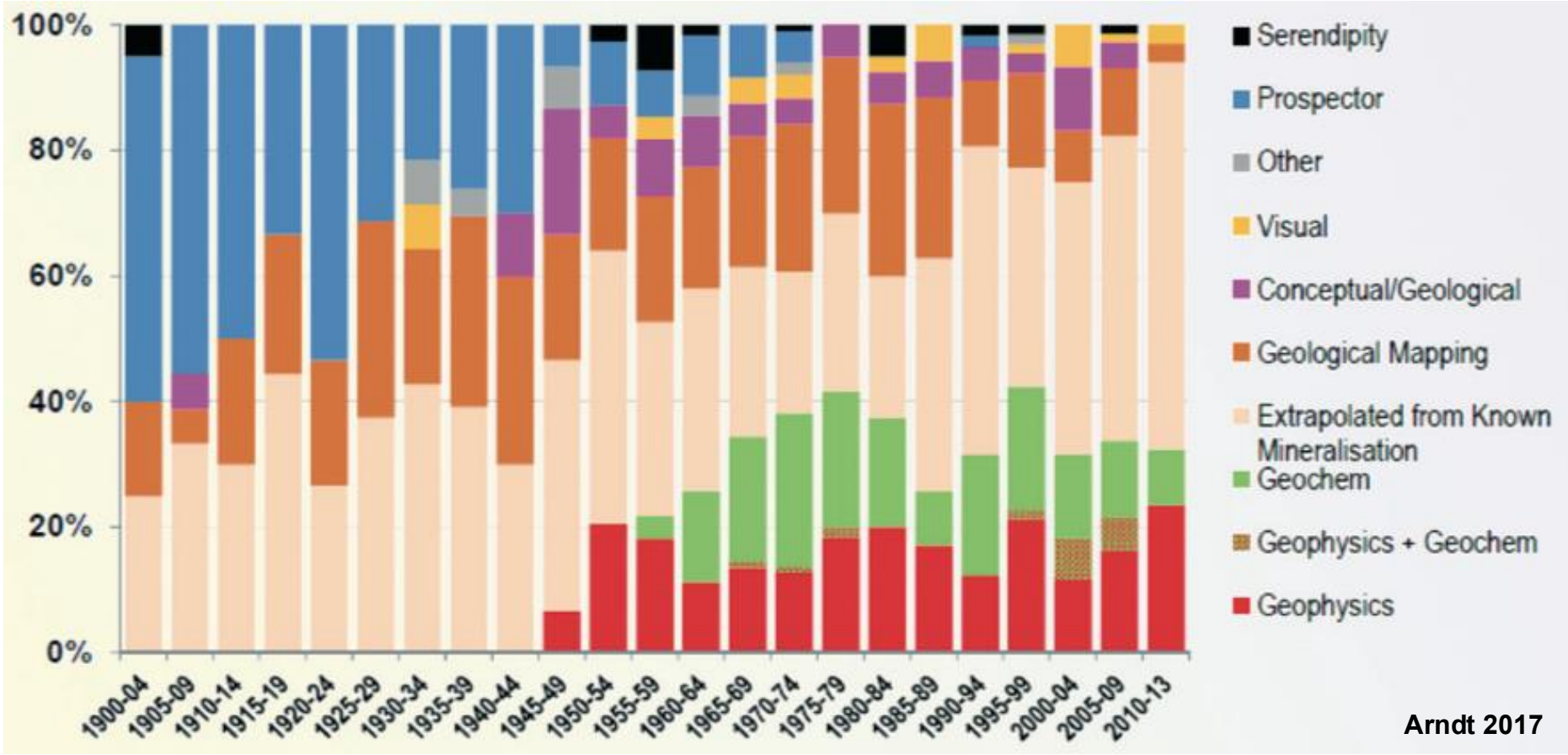
Energy and Resources – Exploration and Production
Production and processing



Pell 2021

Energy and Resources – Exploration and Production
Role of geoscientists and skills

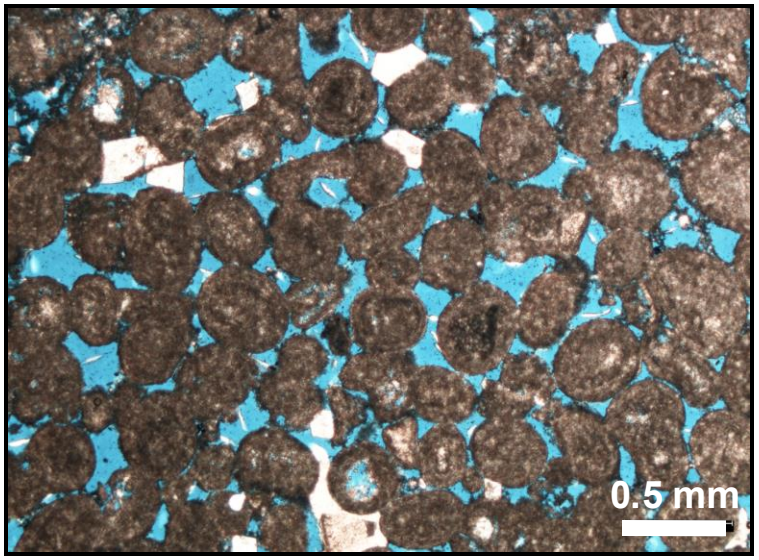
Primary search method for base metal deposits

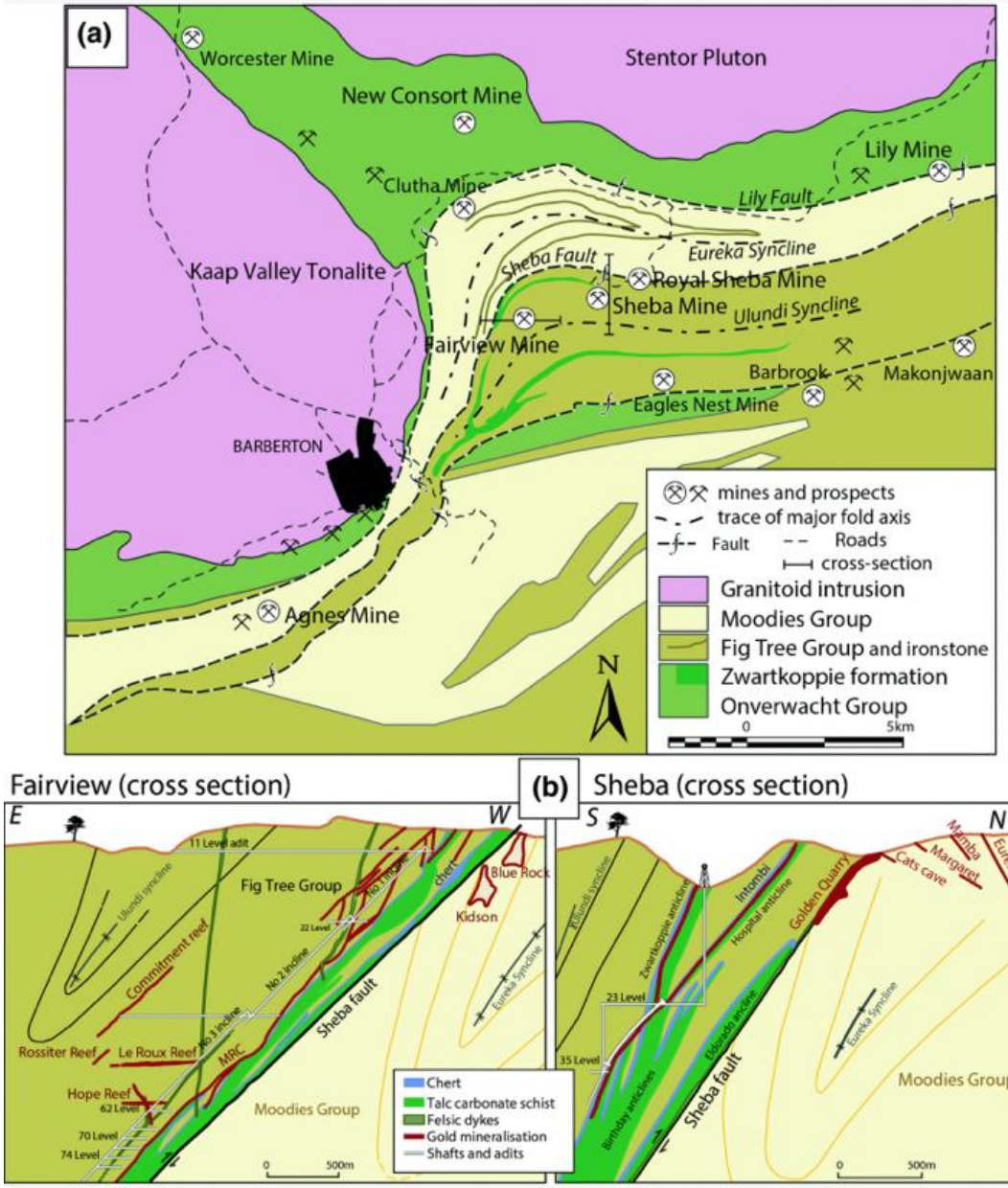


Arndt 2017

Energy and Resources – Exploration and Production
Role of geoscientists and skills

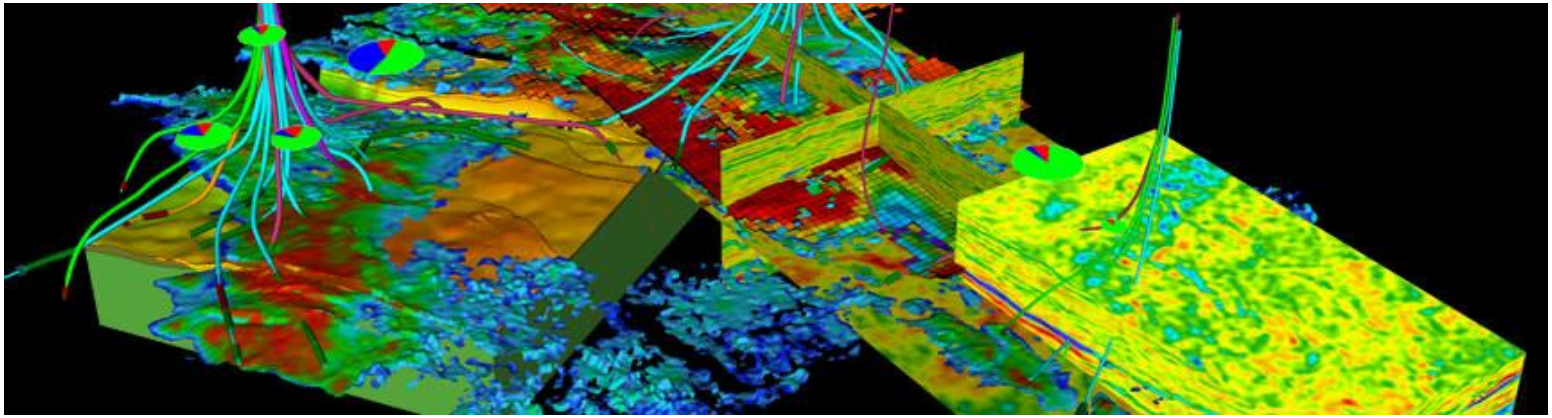
- Mapping at regional and small scale
- 3D visualization
- Observational skills



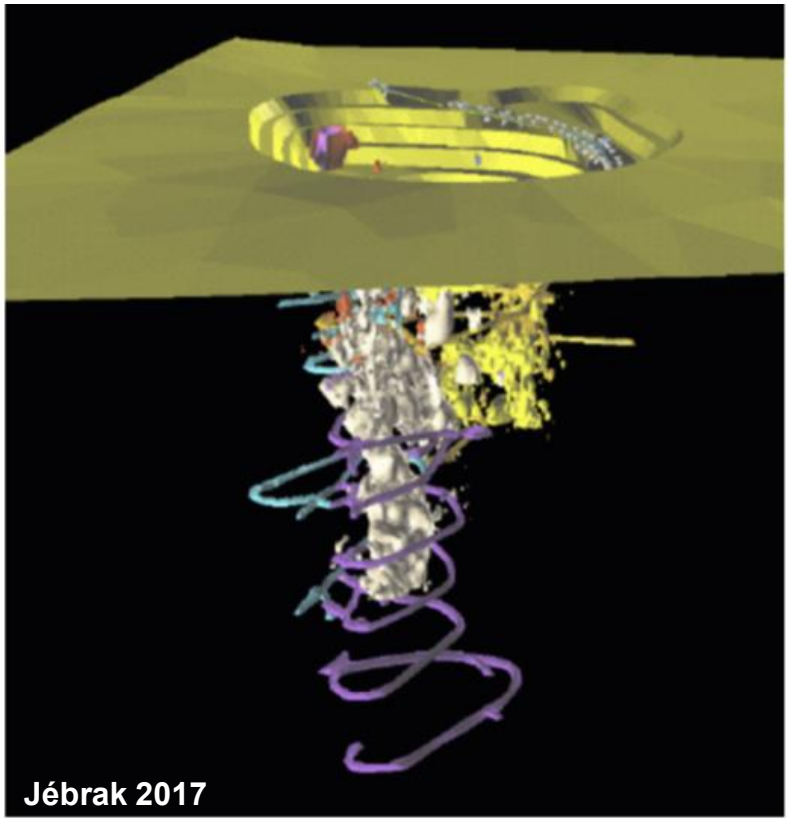


Agagi 2019

Energy and Resources – Exploration and Production
Role of geoscientists and skills

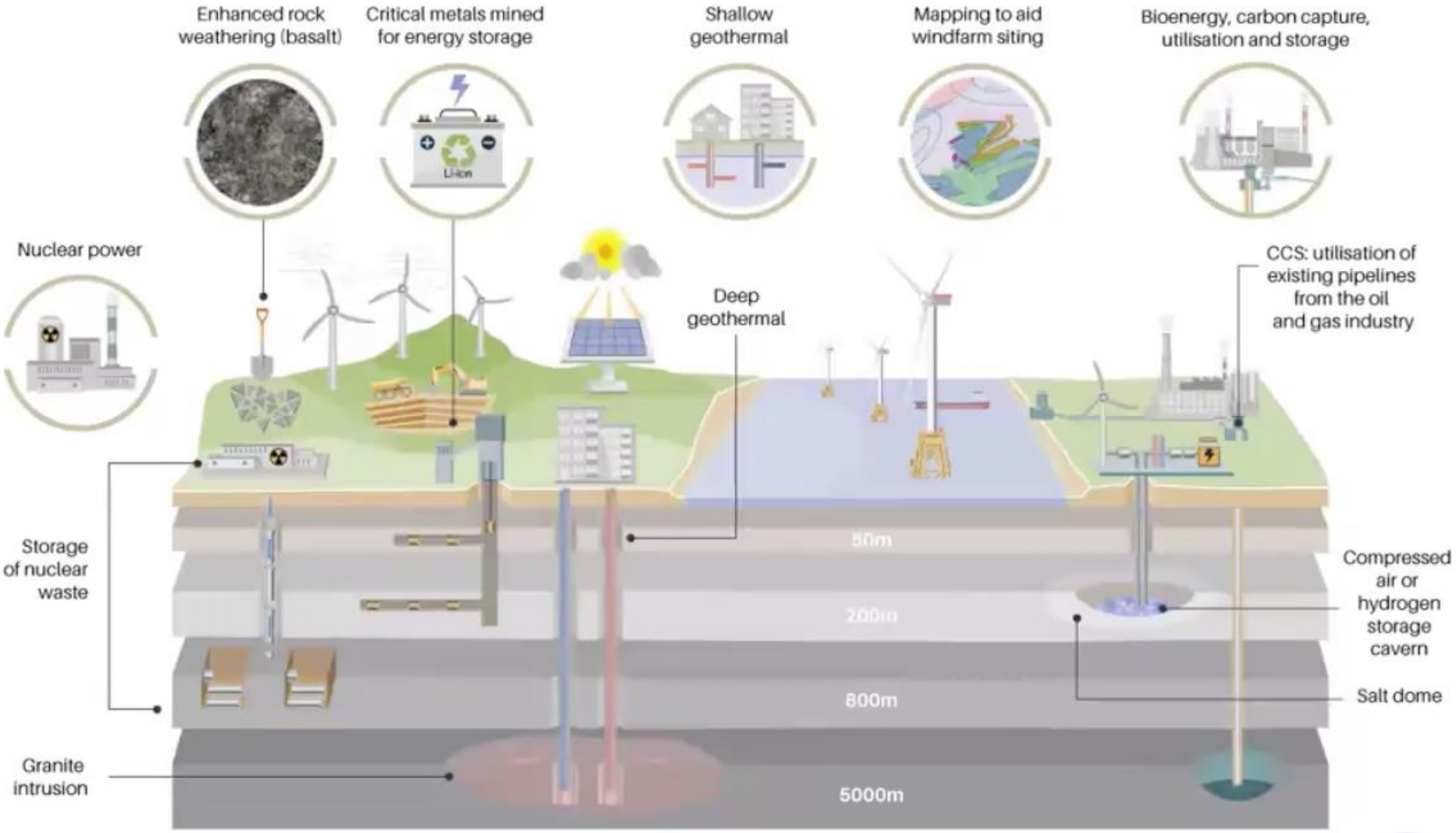


- Numerical skills, use of geo software
- Independence
- Teamwork

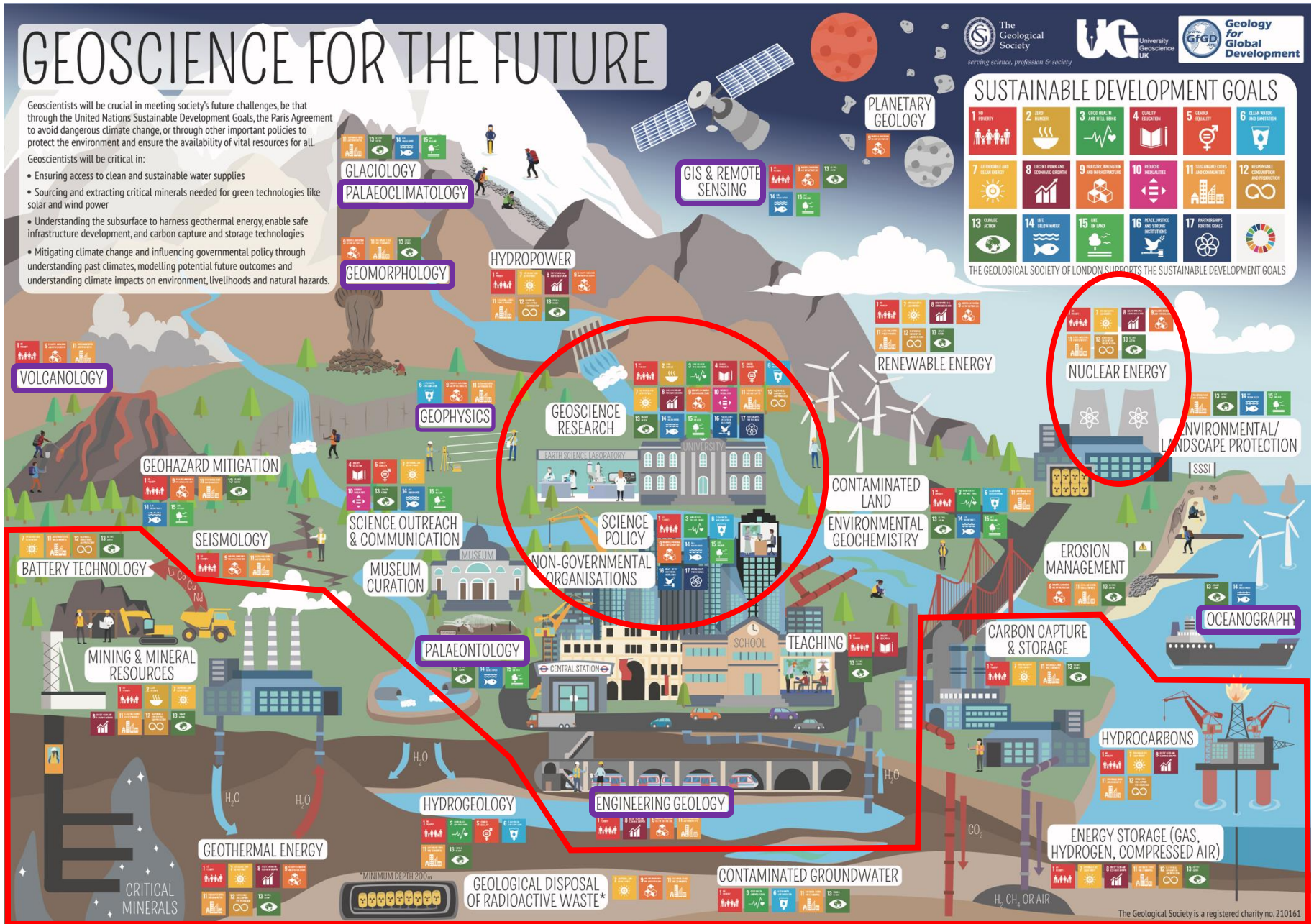


Energy and Resources
Summary week 1

Everything starts with a rock



Energy and Resources
Summary week 1



Energy and Resources
Summary week 1

- Energy & resources include raw materials extracted from the Earth: metallic and non-metallic mineral deposits, as well as fossil fuels and ground water. Renewable resources will become more important in the future, but are not covered in this unit.
- Human society has developed using raw materials, and raw material extraction will remain important to:
  - meet economic needs of developing nations
  - meet requirements of energy transition to low-C economy
- The right combination of geological processes and constraints forms a deposit. Its economical extraction is governed by economic, environmental, legal and societal constraints.
- Exploration using mapping, geophysical and geochemical methods aims to reduce the inherent uncertainty in the subsurface.
- Geoscientists know how to find and assess a deposit using parameters such as mineral grade, volumes in place. Geoscientists are no experts in the economic, environmental, legal and societal factors, but are aware of them. Key skills are:
  - observational skills at a range of scales
  - 3D understanding of the subsurface
  - numerical skills